

TEJARAM CHAUDHARI

+91-9139392550 | ✉tejaschaudhari131@gmail.com | 🌐tejaschaudhari131.github.io
🌐linkedin.com/in/tejaramchaudhari | 🌐github.com/tejaschaudhari131 | 📍Pune, Maharashtra, India

EDUCATION

Trinity College of Engineering & Research (TCOER)

July 2021 - June 2025

B.E. Electronics & Telecommunication Engineering with honors in Cyber Security

Pune, India

- **CGPA:** 8.10 / 10.0
- **Scholarship:** Awarded **Rajarshi Chhatrapati Shahu Maharaj Shikshan Shulkh Shishyavrutti Yojna** (INR 176,000) by the Directorate of Technical Education for pursuing a Bachelor of Engineering degree (2021 - 2025).
- **Key Courses:** Computer Networks, Network Security, Cryptography, Ethical Hacking, Data Analytics, Cloud Computing, Signals and Systems, Embedded Systems & Microcontrollers, Internet of Things (IoT), Control Systems, Project Management

EXPERIENCE

Bhabha Atomic Research Centre (BARC)

Jan 2025 - Mar 2025

Research Intern, Electron Beam Centre

Mumbai, India

- Engineered a high-precision microcontroller-based system using Cypress PSoC 5LP to generate three synchronized pulses, achieving a timing precision of 83.33 ns.
- Implemented Modbus RTU protocol over UART (9600 bps), ensuring compliance with industrial automation standards for remote configuration.
- Developed a local Human-Machine Interface (HMI) using Nextion, allowing real-time parameter adjustments.
- Devised a 'union'-based approach in C firmware to efficiently handle 32-bit float-to-register conversions.

Palp Technologies

Oct 2023 - Nov 2023

Research Intern

Pune, Maharashtra

- Designed an IoT-based smart irrigation system utilizing sensor data automation to optimize agricultural water usage.
- Integrated PLC SCADA automation for real-time monitoring, enhancing water management efficiency.
- Evaluated energy efficiency of Solar Photovoltaic (SPV) pumps across different kW and HP ratings.
- Assessed sustainability impact, aiming to reduce water waste and promote energy-efficient farming.

PROJECTS

Digital Twin for Sensor Network Management (Final Year Project)

Academic Project

Tools: Python, MQTT, TLS, LSTM



- Architected a secure cyber-physical digital twin for IoT sensor networks using MQTT over TLS for encrypted data ingestion.
- Developed a physics-informed model coupled with an LSTM-based model for predictive maintenance.

Adversarial GAN for Intrusion Detection Systems (IDS)

Personal Project

Tools: Python, TensorFlow, CIC-IDS2017



- Developed a GAN framework to generate adversarial network traffic against deep learning-based IDS.
- Evaluated IDS robustness based on evasion rate and accuracy degradation to highlight system vulnerabilities.

F450-Class Autonomous Drone for Surveillance

Academic Project

Tools: Pixhawk, ArduPilot, Mission Planner, Embedded C

- Engineered an autonomous F450-class quadcopter using a Pixhawk flight controller for waypoint-based surveillance missions.
- Configured ArduPilot firmware and implemented sensor fusion for stable flight control and navigation.
- Integrated telemetry and mission planning using Mission Planner for real-time monitoring and control.

Hydrobot: Autonomous Water Cleaning Robot

Hackathon Project

Tools: AI, CNNs, Robotics



- Developed an AI-powered robot for autonomous collection and sorting of aquatic debris using CNNs.
- Conceived and prototyped within a 24-hour hackathon, winning 1st Runner-Up at a national innovation marathon.

PUBLICATIONS

C=CONFERENCE, B=BOOK CHAPTER

[C.1] Aditya Joshi, Pritam Bhalgat, Pratibha Chavan, **Tejaram Chaudhari**, Sumit Patil (2024). **Guarding Against Quantum Threats: A Survey of Post-Quantum Cryptography Standardization, Techniques, and Current Implementations**. In *14th Int. Conf. on Applications and Techniques in Information Security*. (Springer).

[C.2] **Tejaram Chaudhari**, Aditya Joshi, Devkumar Bharti, Pratibha Chavan, M. K. Deshmukh, Pritam Bhalgat (2024). **A Comparative Study of Ensemble Learning Models for Accurate Solar Irradiance Forecasting**. *8th Int. Conf. on Computational Systems and Information Technology*. (IEEE Xplore).

[B.1] M. K. Deshmukh, **Tejaram Chaudhari**, et al. (2025). **Hydrobot: An Autonomous Solution for Surface-Level Aquatic Debris Collection Using CNNs**. In *Frontiers in Computing and Systems (Upcoming Book)*. (Springer).

SKILLS

- **Languages:** Python, C++, PHP, SQL, Bash
- **Hardware & Embedded:** PSoC 5LP, Arduino, Raspberry Pi, Pixhawk Flight Controller, Modbus RTU, UART
- **Tools & Platforms:** Git, MATLAB, Ansys, AWS, Google Cloud Platform (GCP), Linux, SAP Analytics Cloud
- **Core Domains:** Cyber Security, Cryptography, IoT, Embedded Systems, Machine Learning

HONORS AND AWARDS

2nd Runner Up, SAP Code Unnati Innovation Marathon 2.0 <i>SAP (Corporate Sponsor)</i>	March 2024
• Recognized at a highly prestigious corporate-sponsored innovation hackathon (Gujarat Edition).	
2nd Position, IMPACT 2023 (Ideathon) <i>Google Developer Student Club (GDSC)</i>	March 2023
• Recognized for innovative problem-solving in the tech ecosystem.	
1st Rank, Project Competition <i>Trinity College of Engineering and Research</i>	June 2022
• Secured top rank in the college-level project competition.	
Winner, Application Developer’s Day <i>College Technical Fest</i>	Oct 2022
• Winner of the project competition demonstrating coding and development skills.	

LEADERSHIP EXPERIENCE

Founder and Director <i>thinkMINNT Foundation (Section 8 Non-Profit)</i>	Nov 2024 - Present
• Established a non-profit to empower high school students in AI, Cybersecurity, and Data Science.	
• Spearheaded interdisciplinary programs integrating Math, Informatics, and Technology.	
Co-Founder and Webmaster <i>Cybersecurity Club, Trinity College of Engineering and Research (TCOER)</i>	July 2024– May 2025
• Developed and maintained the Cybersecurity Club website; organized workshops and cybersecurity events.	
• Fostered student engagement through hands-on activities in security, privacy, and networking.	
Coordinator <i>6th International Conference on Engineering & Technology (ICET)</i>	April 2025
• Orchestrated logistics for an international research conference, facilitating global research exposure.	
Editor-in-Chief <i>Department & College Newsletters, TCOER</i>	Jan 2024 - Nov 2024
• Led the publication of 11 editions of the department’s technical newsletter, mentoring a 5-member student editorial team in content curation and technical writing.	
• Sourced, edited, and published scholarly articles, faculty achievements, and student research projects for the official college website and monthly newsletters.	
Cloud Co-Lead <i>Google Developer Students Club (GDSC)</i>	Aug 2023 - Aug 2024
• Guided 120+ students through Google Cloud programs and Generative AI pathways.	

CERTIFICATIONS

Google Cloud Career Practitioners Completion	Oct 2022 - Jan 2023
SAP Certified: IoT Cyber Security & SAP Analytics Cloud	2024-25
SAP Certified: Emerging Tech (ML, IoT, Deep Learning, ABAP)	2023-24
Cisco Networking Academy: Networking Basics Certificate	2024

ADDITIONAL INFORMATION

Sports: University Representative for Table Tennis & Chess (Savitribai Phule Pune University, 2023).

Co-Curricular: Participant in AI in Media Innovation Hackathon (2025), National Level Techathon (2024).